Frequentist, Bayesian and machine-learning analysis of the stiffness of resistant breast-cancer cells from AFM force curves

Giacomo Moretti
Università degli Studi di Milano-Bicocca

We analyse 257 AFM force–distance curves measured on the cytoplasm and on the nucleus of two resistant breast-carcinoma lines (MCF7R and T47DR). Each curve is reduced to the effective elastic parameter A of a Hertz-like model, $F = A\,\delta^{3/2}$, and the distributions of $\log A$ are compared between regions with the same statistical toolbox in a frequentist (profile likelihood, toy Monte Carlo, Holm correction, systematics), a Bayesian (MCMC, Bayes factors, hierarchical model) and a machine-learning (anomaly detection with injected defects, classification, mixtures) setting. The nucleus is softer than the cytoplasm in both lines: $\Delta = \mu_{nucleus} - \mu_{cytoplasm} = -0.81 \pm 0.08$ (MCF7R) and -0.25 ± 0.10 (T47DR) in $\log A$, i.e. a ratio of typical A of 0.44 and 0.78. The evidence is overwhelming for MCF7R and only weak for T47DR with the fit parameter (Bayes factor $\sim$ 4:1 despite $p \simeq 0.01$), but a model-independent check on the force at fixed indentation confirms the effect in both lines. The analysis is limited by pseudo-replication and by the coincidence of cell line and measurement session.

I. INTRODUCTION

Cell stiffness measured by atomic force microscopy (AFM) is a label-free marker of cell state, and of the response of tumour cells to treatment. The aim of this work is to compare the mechanical properties of two resistant breast-carcinoma lines, MCF7R and T47DR, in two measurement regions (cytoplasm and nucleus) at 48 h, starting from raw force–distance curves. The emphasis is on the *statistical* treatment: every comparison is made with explicit models, its behaviour is validated with toy Monte Carlo experiments, and the dependence on analysis choices is quantified. The pipeline is implemented in five notebooks (reading and QC, contact point and fit, frequentist, Bayesian and machine-learning analysis; see Sec. VIII).

II. DATA AND CURVE REDUCTION

The data are 257 JPK force files in four groups: MCF7R cytoplasm (55 curves), MCF7R nucleus (60), T47DR cytoplasm (59) and T47DR nucleus (83), measured with an MLCT-BIO triangular cantilever (spring constant $\simeq$ 0.12 N/m). Each file holds an extend (approach) and a retract segment; the extend segment is used for the mechanical fit, the retract one for the adhesion. The curves are the «good curves» already selected by the experimenters; a preliminary instrumental QC (robust z-scores of noise, force excursion and jumps) is computed but *no curve is discarded*.

For each extend curve the baseline is a robust line fitted on the first 30% of the points (the tip is far from the surface), the noise σ_b is the MAD of its residuals, and the contact point is initialised where the smoothed force exceeds $\max(8\sigma_b,\, 5\%F_{max})$ for 20 consecutive points. In the window $0.08F_{max} < F < 0.8F_{max}$ the data are fitted with

$$F(\delta) = A\,\delta^{3/2}, \qquad \delta = z_c - z, \tag{1}$$

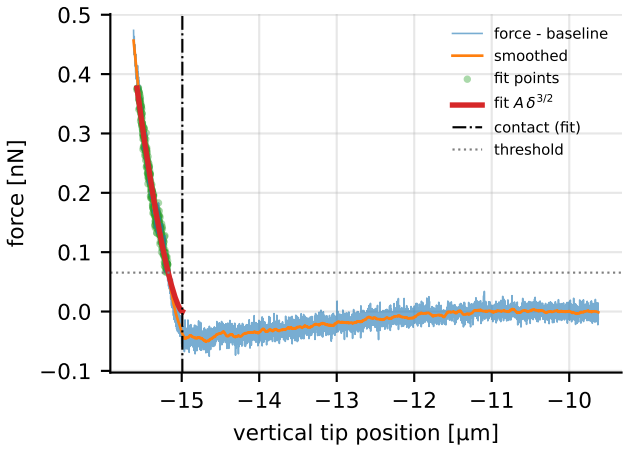


FIG. 1. Example of baseline-corrected curve, contact point and Hertz-like fit (MCF7R cytoplasm).

with A and the contact point z_c free [1]. The tip radius and the Poisson ratio are not available, so A is an *effective* parameter in nN μm$^{-3/2}$ and not the Young modulus; the nominal tip geometry is pyramidal, for which a Sneddon-type law [2] would have exponent 2, but the free exponent measured on the curves has median 1.47, consistent with the 3/2 used here; in any case only *relative* comparisons are meaningful. The fits have median $R^2 = 0.986$ and a relative statistical error on A of 0.8%, far smaller than the spread of A between curves (Sec. IV), which is therefore intrinsic. Figure 1 shows one fit. A is positive and right-skewed, so the analysis uses $\log A$ (with A in nN μm$^{-3/2}$).

III. STATISTICAL MODEL

In each group g we model $x_i = \log A_i \sim \mathcal{N}(\mu_g, \sigma_g^2)$ (a log-normal A). Figure 2 shows the distributions: the Gaussian is adequate for the bulk, and $\log A$ is much

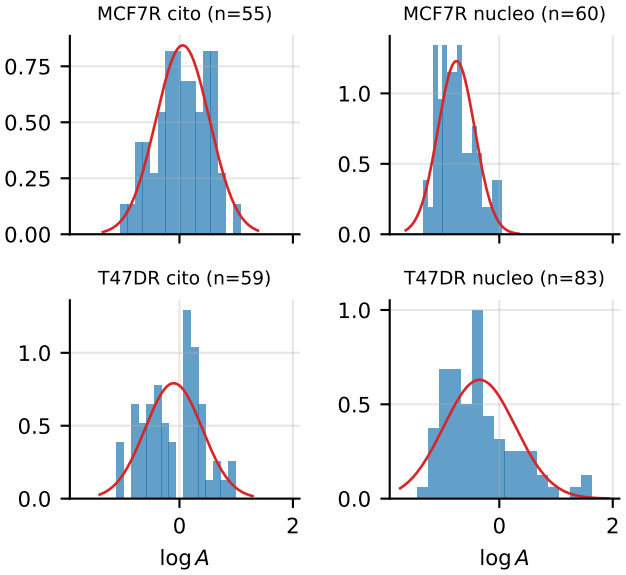


FIG. 2. Distribution of $\log A$ in the four groups with the fitted Gaussian.

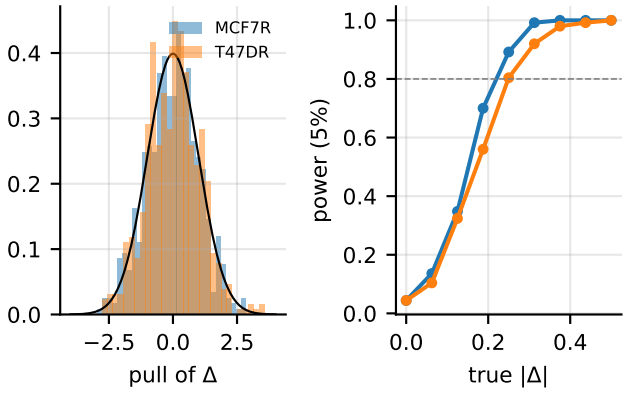


FIG. 3. Toy Monte Carlo of the profile-likelihood estimator. Left: pull of Δ (black: standard normal). Right: power of the Wilks test versus the true $|\Delta|$.

more symmetric than A (skewness of A from 0.8 to 3.1, of $\log A$ from -0.2 to 0.9; the largest values are in the nucleus groups, see Sec. V). The parameter of interest is the nucleus–cytoplasm shift $\Delta_l = \mu_{l,nucleus} - \mu_{l,cytoplasm}$ for each line l; e^{Δ} is the ratio of the typical A, and $\Delta < 0$ means a softer nucleus.

IV. FREQUENTIST ANALYSIS

Estimation. The parameters are obtained from the unbinned likelihood with `iminuit` [3]; Hesse and Minos errors coincide with the analytic $\sigma/\sqrt{n}$ (e.g. MCF7R nucleus: $\hat{\mu} = -0.757 \pm 0.042$, $\hat{\sigma} = 0.325 \pm 0.030$).

Profile likelihood. For fixed Δ the maximisation over the other parameters is analytic up to $\mu_{cytoplasm}$, so the profile is computed exactly. The 68% intervals and the Wilks test [4] give $\Delta_{\mathrm{MCF7R}} = -0.814\,^{+0.077}_{-0.076}$ ($q = 70.7$, $p < 10^{-15}$) and $\Delta_{\mathrm{T47DR}} = -0.247\,^{+0.096}_{-0.095}$ ($q = 6.5$, $p = 0.011$). Welch, Mann–Whitney, permutation and bootstrap-of-medians tests agree (T47DR: $p = 0.011$, 0.003, 0.015; median difference -0.27, 95% interval $[-0.67, -0.07]$), so the conclusion does not depend on the Gaussian assumption.

Toy Monte Carlo validation. Generating 1500 experiments per line from the fitted model, the estimator is unbiased (bias $< 10^{-3}$), the pulls have mean 0.01 and width 0.98–1.00, the 68% intervals cover 0.680 and 0.679, and the size at the nominal 5% is 0.059 (MCF7R) and 0.052 (T47DR), equal to Welch and Mann–Whitney. The power at the real sample size is 1.00 for MCF7R and 0.73 for T47DR; 80% power requires $|\Delta| \gtrsim 0.22$–0.25, so T47DR sits just below the sensitivity threshold (Fig. 3).

Interaction and multiple tests. A nested test of the additive model $\mu_{l,r} = a_l + b_r$ against the full model rejects it ($q = 20.6$, $p = 6 \cdot 10^{-6}$): the nucleus effect depends on the line. Between lines, $\Delta_{\mathrm{T47DR–MCF7R}}$ is -0.16 ± 0.09 in the cytoplasm and $+0.41 \pm 0.08$ in the nucleus; however the two lines were measured on different days, so this comparison is confounded with the session. The four confirmatory tests after Holm correction have $p = 2 \cdot 10^{-16}$, 0.021 (T47DR nucleus–cytoplasm), 0.089 and $5 \cdot 10^{-6}$.

Systematics. Nine variants change one preprocessing choice at a time (baseline fraction, threshold, lower and upper fit-window edge). With the Barlow criterion [5] no variant is significant for MCF7R and the total systematic is 0.07 for T47DR, smaller than its statistical error 0.10 (Fig. 6, right).

V. BAYESIAN ANALYSIS

Model and sampling. The same model with weakly informative priors ($\mu_g \sim \mathcal{N}(0,1)$, $\sigma_g \sim \mathrm{HalfNormal}(1)$) is sampled with `emcee` [6] (40 walkers; $\hat{R}_{max} < 1.03$, integrated autocorrelation time $\lesssim 100$ steps). The posteriors of Δ (median, 68% interval) are $-0.81\,[-0.89, -0.74]$ and $-0.25\,[-0.34, -0.14]$, with $P(\Delta < 0) = 1.00$ and 0.993; they coincide with the profile intervals and are stable under wide, narrow and Jeffreys priors (shifts ≤ 0.03).

Bayes factors. Marginalising σ analytically, the evidence of H_1 ($\Delta \sim \mathcal{N}(0, \tau^2)$) against H_0 ($\Delta = 0$) is computed by exact numerical integration. For MCF7R $\log_{10} BF_{10} \simeq 14$ for any $\tau \gtrsim 0.3$. For T47DR $\log_{10} BF_{10} \leq 0.8$ (about 6:1 at best, 0.6 at $\tau = 0.5$) and negative for $\tau \gtrsim 5$ (Fig. 6, left): a small effect that $p \simeq 0.01$ would call significant is only weak evidence in the Bayesian sense [7].

Model checks. A Student-t likelihood is not preferred (WAIC [8] difference -0.4, $\nu \simeq 39$), so heavy tails are not

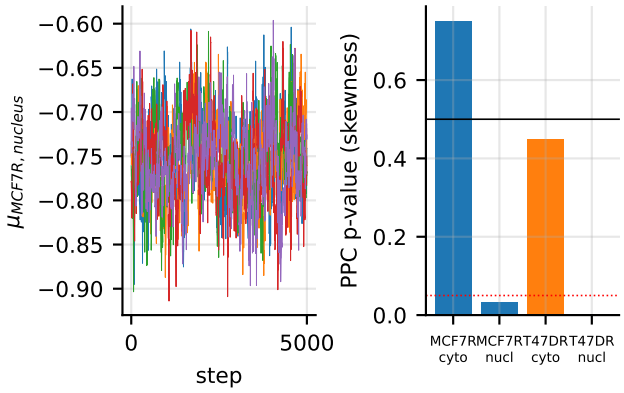


FIG. 4. Left: trace of $\mu_{MCF7R,nucleus}$ for five walkers. Right: posterior predictive p-value of the skewness in each group (dotted: 0.05).

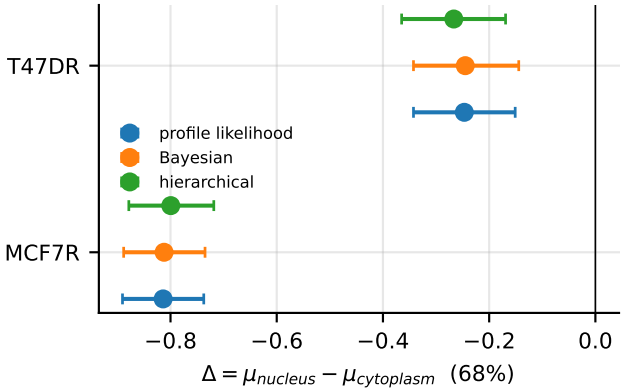


FIG. 5. Δ (nucleus – cytoplasm) in $\log A$ with the three methods.

the problem. The posterior predictive check reproduces mean and standard deviation in all groups but not the *skewness* of the nucleus groups (T47DR nucleus: $p_{PPC} \simeq 0$; MCF7R nucleus: 0.04; Fig. 4). The chains are well mixed (Fig. 4, left).

Hierarchical model. Writing $\mu_{l,nucleus} = a_l + b + u_l$ with $u_l \sim \mathcal{N}(0, \tau_u^2)$, the common effect is $b = -0.49\,[-0.77, -0.16]$ and $\tau_u = 0.43\,[0.26, 0.71]$, confirming that the effect differs between lines. Partial pooling moves the line effects by less than 0.03.

Low statistics. With 5–80 curves per group, the 68% credible interval over-covers slightly (0.72 at $n = 5$, 0.68 at $n = 80$) and the profile interval under-covers (0.62 at $n = 5$, 0.66 at $n = 80$), at the price of intervals longer by 30% at $n = 5$; the two agree for $n \gtrsim 20$.

VI. MACHINE LEARNING

Anomaly detection with known truth. Since all real curves are pre-selected, the efficiency of anomaly detection is measured on 1000 synthetic Hertz-like curves whose parameters are sampled from the real fits (including a correlated model mismatch), with 5% of injected defects (`no_contact`, `saturation`, `huge_noise`, `jump`, `wrong_slope`). Isolation Forest [9] and a robust Mahalanobis distance, both unsupervised and trained only on good curves, reach AUC 0.88 and 0.87; at a 5% false-positive rate the efficiencies are 100% for `no_contact` and `huge_noise` (these already fail the fit), 60–90% for `saturation` and `jump`, and only 10–30% for `wrong_slope` (Fig. 8, left; 10 defects per type, i.e. about ±15%). Applied to the real curves (trained on them, cross-validated), the detector flags 8 curves in common with the second detector, for visual inspection only.

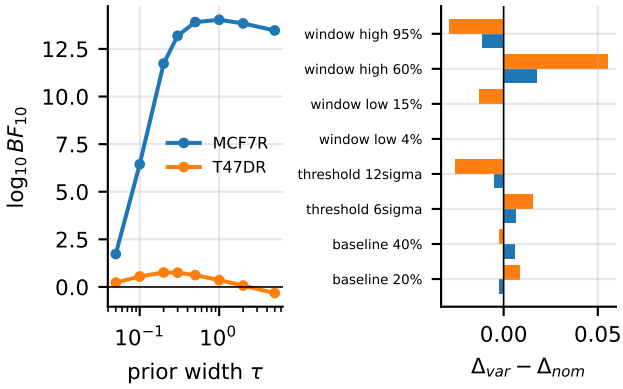


FIG. 6. Left: Bayes factor versus the prior width on Δ. Right: shift of Δ for each preprocessing variant.

Classification of the region. From $\log A$, maximum indentation and two adhesion features, a logistic model separates nucleus and cytoplasm with AUC 0.95 in MCF7R and 0.63 in T47DR (cross-validated; permutation tests $p = 0.005$ and 0.025), consistent with the effects of the previous sections (Figs. 8, right, and 7). Trained on one line and tested on the other, the AUC is 0.66 (MCF7R→T47DR) and 0.96 (T47DR→MCF7R): the sign of the effect is shared, its size is not.

Curve shape and a model-independent check. A sixth notebook resamples every force–indentation curve on 60 points and removes the amplitude. Without labels the curves do not cluster by line (adjusted Rand index ≈ 0) and only weakly by region (≤ 0.2). Classification from the shape alone gives AUC 0.62 in MCF7R (where the amplitude carries the information, 0.95) but 0.76 in T47DR, against 0.63 from the scalar features: in T47DR the shape contains region information that A does not, which needs checking on the raw curves. Because the curves end at a nearly constant setpoint force, the maximum indentation is a function of A, so the strong correlation ($\simeq -0.9$) of $\log A$ with it is expected and not an artifact. As a test independent of the contact model, the force at a fixed indentation of 300 nm is lower in the nucleus in both lines, $\Delta \log F = -0.76$ [95% interval $-0.91, -0.61$] (MCF7R) and -0.35 $[-0.53, -0.18]$

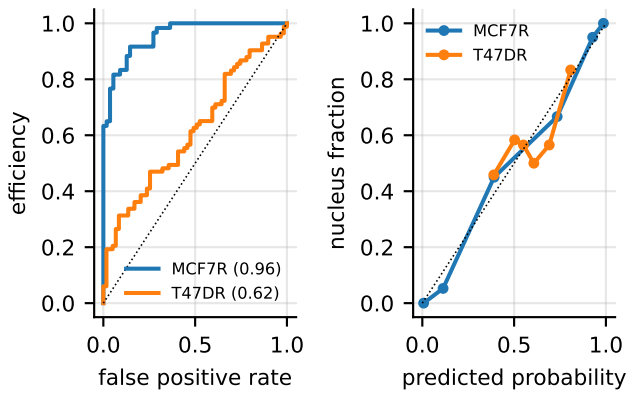


FIG. 7. Cross-validated ROC (AUC in the legend) and calibration of the logistic nucleus/cytoplasm classifier.

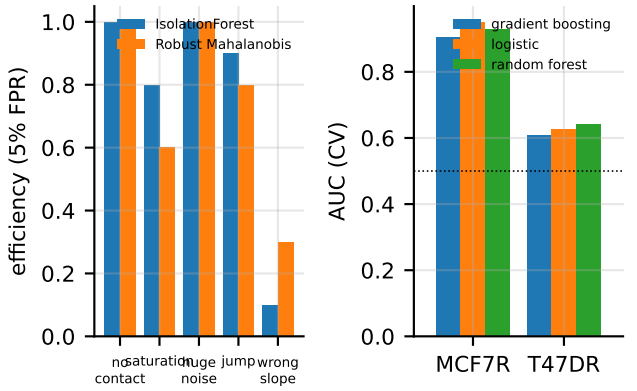


FIG. 8. Left: anomaly-detection efficiency on synthetic defects. Right: cross-validated AUC of the nucleus/cytoplasm classifiers.

(T47DR, $p \approx 2{\cdot}10^{-4}$): the conclusion does not depend on the Hertz fit or its window, and for T47DR it is clearer than with A. Conformal prediction at 90% confidence gives a single label to 79% of the MCF7R curves but only 25% of the T47DR ones, and the learning curve saturates at AUC ≈ 0.95 after about 10 curves per class for MCF7R, while T47DR stays near 0.6.

Mixtures. A parametric-bootstrap likelihood-ratio test favours two Gaussian components only for T47DR nucleus ($p = 0.010$; 71% at $\log A = -0.64$ and 29% at 0.35); this is the origin of the skewness seen in the posterior predictive check.

VII. DISCUSSION AND LIMITS

The three approaches agree on the size of the effect and differ only in how they weigh the evidence for T47DR, which is the honest summary: the nucleus of MCF7R is clearly softer than its cytoplasm ($\times 0.44$ in A), while for T47DR the effect ($\times 0.78$) is indicative and below the 80%-power threshold at the available sample size. The main limits are: (i) **pseudo-replication**: several curves come from the same cell and the cell identifier is not available, so the errors are optimistic; a hierarchical model with cell as grouping factor is the natural extension; (ii) **cell line ≡ session**: the lines were measured on different days (25/02 and 18/03), so differences between lines cannot be separated from session effects; (iii) **pre-selected sample**: no real defects are available, so the efficiency of the anomaly detection is optimistic; (iv) **effective parameter**: A is not the Young modulus and assumes a 3/2 exponent; (v) the high-A tail of T47DR nucleus suggests subpopulations, which a comparison of means does not describe.

VIII. CODE

All results are produced by six notebooks in `Code_General/` (`01` reading and QC, `02` contact point and fit, `03` frequentist, `04` Bayesian, `05` machine learning, `06` curve-shape extensions) and the shared module `afm_tools.py`; the figures are regenerated by `Docs/report/make_figures.py` from `Code_General/results/`. The raw data are not distributed with the repository (see the README).

Appendix A: Notation and formulas

Quantities. A: effective elastic parameter (nN μm$^{-3/2}$); $x = \log A$; $\mu_{l,r}, \sigma_{l,r}$: mean and standard deviation of x for line $l \in \{\text{MCF7R, T47DR}\}$ and region $r \in \{\text{cytoplasm, nucleus}\}$; $\Delta_l = \mu_{l,nucleus} - \mu_{l,cytoplasm}$; e^{Δ}: ratio of typical A.

Profile likelihood. With n_c, n_n curves and $S_c(\mu) = \sum_i (x_i - \mu)^2$ over the cytoplasm curves, $S_n(\mu, \Delta) = \sum_j (x_j - \mu - \Delta)^2$ over the nucleus curves, profiling $\sigma_c^2 = S_c/n_c$ and $\sigma_n^2 = S_n/n_n$ gives

$$-2 \ln \mathcal{L}_{\text{prof}}(\Delta) = \min_{\mu} \left[n_c \ln \frac{S_c(\mu)}{n_c} + n_n \ln \frac{S_n(\mu, \Delta)}{n_n} \right] + \text{const.}, \tag{A1}$$

and the 68% interval is where this rises by one unit above its minimum; the test of $\Delta = 0$ uses $q = -2 \ln \mathcal{L}_{\text{prof}}(0) + 2 \ln \mathcal{L}_{\text{prof}}(\hat{\Delta}) \sim \chi^2_1$.

Bayes factor. With the improper prior $p(\sigma) \propto 1/\sigma$ common to both hypotheses, integrating σ_c, σ_n out gives

$$p(\text{data} \mid \mu, \Delta) \propto S_c(\mu)^{-n_c/2}\, S_n(\mu, \Delta)^{-n_n/2}, \tag{A2}$$

and $BF_{10} = \iint p(\text{data} \mid \mu, \Delta)\, \mathcal{N}(\mu; 0, 1)\, \mathcal{N}(\Delta; 0, \tau^2)\, d\mu\, d\Delta \,/ \int p(\text{data} \mid \mu, 0)\, \mathcal{N}(\mu; 0, 1)\, d\mu$, evaluated on a grid.

Barlow criterion. A variant with estimates Δ_v and error σ_v gives a significant shift if $|\Delta_v - \Delta_{nom}| / \sqrt{|\sigma_v^2 - \sigma_{nom}^2|} > 2$.

Anomaly score and efficiency. The anomaly score is minus the Isolation Forest score (or the robust Mahalanobis distance); the threshold is the 95th percentile of the cross-validated scores of good training curves, so the false-positive rate is 5% by construction; the efficiency of a defect type is the fraction of defective curves above threshold, counting failed fits as detected.

[1] H. Hertz, Über die berührung fester elastischer körper, Journal für die reine und angewandte Mathematik **92**, 156 (1882).

[2] I. N. Sneddon, The relation between load and penetration in the axisymmetric Boussinesq problem for a punch of arbitrary profile, International Journal of Engineering Science **3**, 47 (1965).

[3] F. James and M. Roos, MINUIT: a system for function minimization and analysis of the parameter errors and correlations, Computer Physics Communications **10**, 343 (1975).

[4] S. S. Wilks, The large-sample distribution of the likelihood ratio for testing composite hypotheses, Annals of Mathematical Statistics **9**, 60 (1938).

[5] R. Barlow, Systematic errors: facts and fictions, arXiv:hep-ex/0207026 (2002).

[6] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, emcee: The MCMC hammer, Publications of the Astronomical Society of the Pacific **125**, 306 (2013).

[7] R. E. Kass and A. E. Raftery, Bayes factors, Journal of the American Statistical Association **90**, 773 (1995).

[8] S. Watanabe, Asymptotic equivalence of Bayes cross validation and widely applicable information criterion in singular learning theory, Journal of Machine Learning Research **11**, 3571 (2010).

[9] F. T. Liu, K. M. Ting, and Z.-H. Zhou, Isolation forest, Proceedings of the 8th IEEE International Conference on Data Mining , 413 (2008).